



*"Transforming the way
we think about water"*

AquaBot

CONCEPT NOTE

Introduction

Access to safe drinking water is a basic human right. The United Nations recognized this right in 2010 due to the harmful effects that poor water quality has on overall human health.

Poor water and sanitation are a major cause of health problems in developing countries, especially for children. In Kenya, the constitution guarantees the right to clean and safe water for every citizen.

Element Access International has developed an automated water kiosk (AquaBot) to address these issues. The AquaBot has provided clean and safe water to **over 10,000 people** across Kenya so far. The AquaBot is integrated with a water treatment system killing all pathogens that cause water borne disease. Additionally, the AquaBot sensitizes and educates rural communities about safe water and hygiene practices.

The AquaBot contributes to achieving Sustainable Development Goal 6, target 6.1.



Over the years, there has been increased funding for water projects in different parts of Kenya. However, these actions have not been effective because often when a water project is developed in a community, it becomes dysfunctional after a few years. Research shows that community water projects become dysfunctional due to the following reasons:

- **Poor sustainability measures.**
- **Community dependency.**
- **Community management problems.**
- **Lack of community willingness to pay for water**
- **Lack of community participation at the beginning of the project.**

To solve these challenges, new approaches are needed for rural water projects in Kenya. Element Access International has developed a sustainable solution known as the AquaBot.

AquaBot Overview

The AquaBot is a fully automated and solar-powered water kiosk. It provides push button operation, easily dispensing an accurate amount of water to each user at an affordable rate. This benefits the community by sustaining the operation for many years into the future. Each AquaBot is placed in a location that benefits the maximum number of people and is installed in conjunction with a deep borehole, ensuring a constant supply of water.

The project follows key principles of sustainable WASH to enhance ownership by engaging the community throughout the implementation and bringing awareness of water safety, hygiene, and sanitation. With every AquaBot we make sure the community understands a pay-for-use water system, and that money generated from the AquaBot will be used for maintenance, repair, and overall sustainability of the project.

The goal of the Aquabot is to make clean and safe water accessible to all in a sustainable way that lasts for many years into the future.

AquaBot Success Stories

NGARENGIRO, LAIKIPIA AQUABOT LAUNCH

The project focused on providing a solution to the water challenges faced by the village of Dipatas in Ngarengiro, Laikipia County. The village's main water source was a seasonal river, which did not provide safe drinking water. Through the AquaBot project, over 2600 people gained access to safe drinking water. Additionally, the project aimed to contribute to the progressive realization of the human right to Water, Sanitation, and Hygiene.



Testimonials from the Community



My name is **Paul Maridadi** and I'm a Community Health Promoter from Laikipia County, located in Ngarengiro. As the CHP I have witnessed the water challenges faced by my community. The available water was a river which was dependable to both human livestock. The river is seasonal hence during dry seasons women and children walked 2 to 3 kilometers to get unsafe water, an outbreak of water-related diseases were rampant, leading to children being unable to attend school. However, since the AquaBot was developed in the our village the incidences of water-related diseases have significantly reduced. The AquaBot provides accessible safe water for drinking and domestic use, greatly improving the health and well-being of our community

My name is **Peter Kilesi** a community member from Leruk village. In my entire life, I have seen a lot of water projects being developed in my community but the challenge has been the same poor management which leads to nonfunctional water projects currently we have more than 3 water projects in our village which are unfunctional. Speaking on behalf of my community we were looking for solutions that would solve the biggest challenge we have with our water projects fortunately that's when we came to know about the AquaBot. We believe the AquaBot design and implementation is what the whole rural Community in Kenya and Africa needs. We are lucky to be among the first village to enjoy the benefit of the AquaBot.





We have completed AquaBot projects in some of the most demanding and challenging rural communities in Kenya. Providing clean and safe water to **over 10,000 people**.



Our solution is developed, and we have established proof of concept.



Our goal is to scale up the AquaBot project to major parts of Kenya and Africa at large.



Our target is to build 100 AquaBots in the next 5 years.



Contact us today for collaboration and/or partnership.

Pictorial



COMPLETE AQUABOT



COMMUNITY LAUNCH GROUP PHOTO



Thank You



ELEMENT ACCESS INTERNATIONAL
"Global Excellence & Innovative Solutions"



KCC Road, Opp. Trojan Depot,
Industrial Area
P. O. Box 2054-10400, Nanyuki



+254 112 902 872



info@elementaccess.org
www.elementaccess.org